THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The loudness of a sound depends mainly on its: 1

(a) speed **(b)** amplitude **(c)** frequency **(d)** wavelength

2. A cell without a well-defined nucleus is a: 1

(a) eukaryotic cell **(b)** animal cell **(c)** plant cell **(d)** prokaryotic cell

3. The tissue that makes the husk of a coconut is the: 1

(a) phloem **(b)** collenchyma **(c)** parenchyma **(d)** sclerenchyma

4. The simplest plants with a thallus-like body are: 1

(a) algae **(b)** gymnosperms **(c)** ferns **(d)** mosses

5. The male reproductive part of a flower is the: 1

(a) petal **(b)** carpel **(c)** sepal **(d)** stamen

6. The surface that reflects the most solar radiation is: 1

(a) fresh snow **(b)** ocean **(c)** dark soil **(d)** forest

7. A 'law' of science and a 'theory' differ because: 1

(a) they are the same
(b) a law is a guess
(c) a theory is untested
(d) a law describes what happens while a theory explains why

8. In paper chromatography the pencil line is drawn at the bottom because: 1

(a) pencil does not dissolve in the solvent **(b)** pencil is faster
(c) it looks neat **(d)** ink is better

9. An anion is a: 1

(a) neutral particle **(b)** negatively charged ion
(c) molecule **(d)** positively charged ion

10. The valency of an element with 7 valence electrons is: 1

(a) 8 **(b)** 7 **(c)** 1 **(d)** 6

11. The equation v squared = u squared + 2as relates: 1

(a) force and mass **(b)** velocity and time only
(c) velocity, acceleration and displacement **(d)** work and time

12. The weight of a body on the Moon compared to that on Earth is about: 1

(a) the same **(b)** one-sixth **(c)** zero **(d)** six times

13. A simple machine is a device that: 1

(a) makes work easier by changing the magnitude or direction of force
(b) does more work than the input
(c) creates energy
(d) needs no effort

14. The minimum distance of the reflecting surface to hear an echo in air (v = 344 m/s) is about: 1

(a) 34.4 m **(b)** 1.72 m **(c)** 17.2 m **(d)** 344 m

15. The endoplasmic reticulum with ribosomes attached is the: 1

(a) vacuole **(b)** rough ER **(c)** smooth ER **(d)** Golgi

16. Animal tissue that covers or lines the body surfaces is: 1

(a) nervous tissue **(b)** muscular tissue
(c) epithelial tissue **(d)** connective tissue

17. Sponges belong to the phylum: 1

(a) Cnidaria **(b)** Annelida **(c)** Mollusca **(d)** Porifera

18. After fertilisation, the ovary develops into a: 1

(a) anther **(b)** fruit **(c)** flower **(d)** seed

19. The ozone layer is present in the: 1

(a) geosphere **(b)** mesosphere **(c)** stratosphere **(d)** troposphere

20. Chemists draw atoms as spheres and bonds as sticks. This is an example of: 1

(a) a unit **(b)** an estimate **(c)** a law **(d)** a model

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Define work in the scientific sense. Give its SI unit. 2

22. How is sound produced? Name the medium needed for its propagation. 2

23. Who discovered the cell? What did he observe? 2

24. What is a tissue? Why do multicellular organisms need tissues? 2

25. What is biodiversity? Why is classification necessary? 2

26. Define reproduction. Why is it important? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A raisin is placed in water and another in a strong sugar solution. Predict and explain what happens. 3

28. Explain with an activity how roots grow at their tips. 3

29. Name any four characteristics used for classifying organisms. 3

30. Describe the binary fission in Amoeba with the steps. 3

31. Explain how the four spheres interact with an example of a rainy day. 3

OR

How does the incident of the aircraft running out of fuel show the importance of units? 3

32. How much solute is needed to prepare 250 g of a 12 per cent (m/m) sugar solution? 3

OR

State three properties of ionic compounds. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Describe the meristematic tissues: types, location and functions with the way they differ from permanent tissues. 5

34. Explain the five-kingdom classification with the basis, characters and examples of each kingdom. 5

35. Describe the different methods of asexual reproduction in animals and plants with examples. 5

36. Explain the Earth as a system, describing the four spheres, the flow of energy and the cycling of matter. 5

OR

Explain, with examples, why measurements are never perfectly exact, and how errors can be reduced. 5

37. Describe the working and use of paper chromatography with the diagram described in words, including R_f ideas. 5

OR

A compound contains carbon and oxygen in the ratio 3 : 8. Find the mass of the compound formed when 18 g carbon is burnt and the mass of oxygen needed. Name the compound. 5

38. Explain how the atomic number and the mass number identify an atom. Complete the table for $^{39}_{19}\mathrm{K}$, $^{32}_{16}\mathrm{S}$ and $^{19}_{9}\mathrm{F}$. 5

OR

Explain uniform circular motion with the tangent velocity direction, the relation v = 2 pi r/T, and calculate the speed of a body at the tip of a 0.5 m radius wheel with period 2 s (pi = 3.14). 5

END OF QUESTION PAPER — Total: 80 Marks

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